

Manuel Magana

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Education

Northeastern University | *M.S. Computer Science*

May 2026

Relevant Courses: Algorithms, Distributed Systems, Computer Networking, Software Engineering, Deep Learning

University of California, Berkeley | *B.A. Molecular and Cell Biology: Neurobiology*

May 2021

Skills

Languages: Python, Go, Java, C, TypeScript, SQL, R, Bash

Frameworks & Tools: FastAPI, React, Next.js, pytest, JUnit, Cucumber, Docker, Amazon Web Services (AWS), Git, Redis, MongoDB, PostgreSQL, DuckDB, Gradle, GitHub Actions, CI/CD

Libraries: pandas, PySpark, scikit-learn, PyTorch, TensorFlow/Keras

Work Experience

Genentech

San Francisco, CA

Bioinformatics Scientist

Nov 2022 – Jul 2025

- Co-designed and deployed a Python validation library with 2 engineers, containerized with Docker; reduced manual review from days to hours for 20+ researchers.
- Processed a 200GB+ mass spectrometry dataset using PySpark after it exceeded pandas' memory limits; converted raw data to Parquet for faster reads, enabling filtering and summarization at scale.
- Co-developed interactive dashboards used by 3+ research groups for experiment tracking and QC review.

Endless West

San Francisco, CA

Junior Data Scientist

Mar 2021 – Nov 2021

- Trained ML pipelines for sensory attribute prediction from chromatography data, with model outputs used to prioritize R&D synthesis targets.
- Developed Python web scrapers to extract and structure data from public databases, replacing hours of manual lookup with a searchable local dataset.

Projects

PayPath | *Go, Next.js, React, MongoDB, Docker, OpenAI API* | [Live Demo](#) | [GitHub](#)

Jun 2026 – Present

- Built and deployed a full-stack personal finance dashboard: a Go REST API (34 endpoints, JWT auth, MongoDB Atlas) and a Next.js/React frontend tracking income, expenses, debts, and net worth, with 5 OpenAI-backed insight reports.
- Engineered a reporting service computing federal/state/FICA taxes, debt payoff timelines, and 90-day cash-flow projections that honor per-occurrence calendar edits (moved bills, one-time purchases, paycheck overrides).
- Cut redundant database load with an in-memory TTL cache and singleflight read-collapsing; per-page bundle endpoints serve each dashboard view in a single round trip.

DADA-85 | *TypeScript, React, Three.js, GLSL, Vitest* | [Live Demo](#) | [GitHub](#)

Jul 2026

- Built an interactive 3D web app rendering a live DOM calculator inside a WebGL room: dual WebGL/CSS3D scenes share one camera, with invisible depth-writing quads solving mesh-over-DOM occlusion; custom GLSL flame shaders.
- Designed a config-driven engine where basic, scientific, and programmer modes (live HEX/DEC/BIN conversion, bitwise ops) are pure configs over one generic core; verified with 84 Vitest + React Testing Library tests.

Trak | *Java, Swing, DuckDB, Redis, MongoDB, Gradle* | [GitHub](#)

May 2026 – Present

- Built a full-stack sprint planning tool in Java with a desktop GUI, CLI, and REST API (21 endpoints), featuring Google OAuth sign-in, email-based password recovery, and a first-run setup wizard.
- Implemented 5 interchangeable database backends (DuckDB, Redis, MongoDB, Parquet, JSON) that users can switch at runtime without code changes; shipped v1.0.0 as a native macOS installer.
- Wrote 254 automated tests (Cucumber BDD + JUnit) covering all 5 database backends, authentication flows, and API endpoints.

WhatToDo | *Python, FastAPI, Docker, AWS, PostgreSQL* | [Live Demo](#) | [GitHub](#)

Jan 2026 – May 2026

- Owned backend design for a 4-person team: a 3-model SQLAlchemy schema on AWS RDS with JWT auth and bcrypt hashing.
- Engineered a dual-provider AI engine (Claude + GPT-4o) with automatic failover between providers, keeping recommendations available even when one provider is down.
- Set up CI/CD with GitHub Actions and deployed containerized API to AWS (App Runner, ECR); achieved 96% test coverage.